



Nutritional Considerations for Concurrent Training

16

Timothy Etheridge and Philip J. Atherton

Introduction

Although not a new training paradigm [1], the concept of concurrent resistance and endurance training has recently gained increasing attention amongst athletes, coaches and exercise scientists alike [2]. The goals of concurrent training are straightforward: increase muscle mass and strength whilst simultaneously increasing aerobic capacity. The key limitation of concurrent training however (discussed in detail elsewhere in this book), is the reported ‘interference effect’ whereby performing endurance exercise (EE) concomitantly with resistance exercise (RE) may inhibit chronic strength and possibly mass gains versus resistance training alone, despite comparable aerobic adaptation to isolated endurance training [1]. Thus, nutritional strategies for the concurrent athlete should aim to:

1. Optimise the post-exercise anabolic environment in an attempt to rejuvenate chronic muscle strength and hypertrophic adaptation whilst;
2. Continuing to promote aerobic adaptation in response to the metabolic stress associated with endurance training.

Detailed examination of the nutritional requirements posed by the unique demands of concurrent training is lacking. Nonetheless, considerable theoretical evidence may be derived from acute and chronic intervention studies investigating the ergogenic influence of nutrient intake upon isolated resistance or endurance exercise/training—which when combined, can be assimilated to concurrent training. Within this

T. Etheridge
University of Exeter, Exeter, UK

P. J. Atherton (✉)
University of Nottingham, Nottingham, UK
e-mail: Philip.Atherton@nottingham.ac.uk

context, this chapter will evaluate the role of established and emerging nutrients in augmenting muscle strength and aerobic adaptations to training. This knowledge will be consolidated to extrapolate practical recommendations that might benefit the concurrent athlete, namely minimising the ‘interference’ effect.

Macronutrients for Supporting Concurrent Exercise Adaptation

Protein Supplementation and Resistance Exercise

Adequate dietary protein intake is an essential component of the exercise x nutrient interaction. RE alone in the fasted state causes a transient (2–3 h post-exercise) rise in both the breakdown and synthesis (i.e. ‘turnover’) of muscle proteins [3]. However, in the absence of protein nutrition the post-RE rise in muscle protein breakdown (MPB) exceeds that of muscle protein synthesis (MPS) [4, 5]; the ensuing negative net muscle protein balance, if continued throughout training, would lead to maladaptation due to failed muscle growth and remodelling responses. Dietary protein intake reverses this by amplifying and prolonging the protein synthetic response to RE to achieve a positive net protein balance [5]. Indeed, when ingested in close proximity to acute RE, a single protein bolus stimulates MPS (regarded as the primary driver of post-RE anabolism) for ~48 h [5]. This is due to enhanced sensitivity of active muscle to the anabolic properties of dietary protein for at least 24 h post-RE [6]. In turn, cumulative periods of extended muscle protein accretion with repeated bouts of RE plus protein feeding ultimately drives training-induced muscle remodelling and hypertrophy [7]. In the context of concurrent RE followed by EE, MPS of contractile myofibrillar proteins is increased in the presence of post-exercise protein administration [8, 9]. Since this ‘normal’ acute anabolic response to concurrent exercise and feeding should translate to comparable adaptive hypertrophy to fed-state RE alone, this finding is somewhat contradictory to the interference effect, the cause(s) of which remain to be defined. Regardless, it is prudent to incorporate protein into the concurrent athlete’s dietary plan. Though what type of protein should be ingested, how much, and how often in relation to concurrent training bouts?

First we have to look at general relationships between dietary protein and muscle protein metabolism. It is well established that of the 20 protein encoding amino acids that comprise whole protein feeding, the 9 essential amino acids (EAA, i.e. those that cannot be synthesised *de novo* and must therefore be obtained through the diet) are responsible for stimulating MPS [10] and indirectly inhibiting MPB by serving as an insulin secretagogue [11]. Of the EAA, the branched-chain amino acids (BCAA), and in particular leucine, provides the most potent anabolic stimulus [12, 13]. The protein synthetic response to whole protein and EAA are also dose-dependent: maximal stimulation of MPS occurs with ~10 g of EAA, equivalent to ~20 g of high-quality (e.g. whey) protein, or ~0.25 g kg⁻¹ body mass [14, 15]. Thus, in the non-training healthy adult ~0.8 g protein kg⁻¹ day⁻¹ is sufficient to maintain

protein balance and muscle mass. Importantly for the athlete, this dose-dependency remains at a saturable 20 g when protein feeding is performed in conjunction with RE [14] above which amino acid oxidation is increased and excess protein catabolised. Thus, simply increasing the protein or EAA load of individual meals will not increase net muscle growth, even in resistance-trained athletes.

Optimising timings of protein intake (i.e. prior to, during or after RE) has received considerable attention for maximising post-RE anabolism and ensuing adaptation. Little evidence exists to support the use of protein supplementation prior to [16] or during RE [17]. Pre-exercise feeding strategies do increase MPS after RE [16], presumably due to blood EAA levels remaining elevated for ~3 h post-feeding. However, MPS is inhibited during exercise because the energy-expensive process of protein synthesis (e.g. via ATP-dependent peptide bonding) is subdued in favour of diverting towards energy producing pathways to fuel muscle contraction [17]. There is therefore likely no adaptive advantage to be gained by proximity-dependent feeding patterns, at least for RE lasting ≤ 1 h. Conversely, the 0–4 h post-RE window represents an indisputable time frame for protein-stimulated muscle growth. Because a single bout of RE sensitises muscle to feeding for at least 24 h, the precise timing of protein intake is likely to be irrelevant to the concurrent athlete, providing protein requirements per se are met. Indeed, despite reports that the ‘anabolic window’ may be truncated to ≤ 4 h in resistance-trained individuals [18], muscle strength and mass gains are similar irrespective of the timing of protein intake [19, 20].

Perhaps more relevant to maximising RE-mediated muscle adaptation is the refractory period in MPS that accompanies individual meals. In response to a single protein bolus, MPS increases for ~1.5 h before returning to baseline by 2 h [21], despite continued availability of EAA in the muscle and circulation. Thus, muscle exhibits a post-feeding period whereby muscle is ‘full’ of AAs, and provision of additional protein will fail to re-stimulate MPS. The same is also true for exercise, where 20 g of high-quality protein acutely maximises post-RE MPS [14]. Sufficient time must therefore elapse between feeding doses in order to avoid ‘wasted’ protein intake during the refractory period. Although the precise duration of this latent period remains to be defined, it is probable that protein feeding every 3–4 h (approximately double the acute period of MPS stimulation) is near optimal. Advocating this strategy is the finding that feeding 20 g protein every 3 h over 12 h led to greater daily muscle protein accretion versus 10 g administered every 1.5 h or 40 g every 6 h [22]. Finally, pre-sleep protein intake should be considered since overnight is the longest fasting and, therefore, catabolic period of the diurnal cycle. Therefore, to offset nighttime catabolic losses of muscle protein and maximise RE-induced remodelling, a protein bolus close to sleeping should be consumed [23].

Protein Supplementation and Endurance Exercise

Examination of the relationship between exercise and protein has historically focussed on RE and augmenting hypertrophic adaptation, primarily via the stimulation of myofibrillar (i.e. contractile proteins) MPS. Nonetheless, acute EE in

isolation also stimulates *mixed* MPS, i.e. representing all muscle proteins, including myofibrillar, mitochondrial and sarcoplasmic fractions [24]. Subsequent studies demonstrated the impressive adaptive specificity of muscle, whereby EE selectively increases mitochondrial but not myofibrillar MPS, whilst the reverse is true following RE [25]. Although a more generic MPS response may occur in untrained individuals [25, 26], the training-related selective activation of mitochondrial biogenesis by EE would confer a phenotypic advantage since mitochondrial expansion drives aerobic adaptation to endurance training (whilst also explaining why EE alone fails to increase muscle size). So how can supplemental protein affect these unique responses to EE? Recent evidence indicates that, similarly to post-RE, it is the contractile myofibrillar proteins that are responsive to amino acids after both endurance [27] and concurrent [8] exercise bouts, with no potentiation of mitochondrial protein synthesis [8, 28]. Thus, protein ingestion does not stand to benefit aerobic adaptation e.g. augmenting maximal aerobic capacity ($\text{VO}_{2\text{max}}$) or rightward-shifting the exercise intensity at 'lactate threshold'. Nonetheless, optimal protein doses during [29] or post-EE [30] might still be advantageous for concurrent training by augmenting daily net muscle myofibrillar remodelling/accrual, which could theoretically minimise the interference effect of endurance training on strength training adaptations.

Summary of Protein Requirements for the Concurrent Athlete

For the essential role of exogenous protein in supporting muscle remodelling after both RE and EE and, indeed, concurrent exercise [8], increased dietary protein or EAA-enriched diets are recommended. Based on the considerations outlined above, per feed the concurrent athlete should aim to consume ~20 g high-quality protein. To achieve the maximum number of anabolic phases within a diurnal cycle, but within the limitation of post-feeding refractory periods, protein boluses should be consumed every 3–4 h irrespective of proximity to training sessions. Additional pre-bedtime protein feeding will also help offset nighttime catabolic losses. Taken together, this equates to an increased (versus recommendations for sedentary individuals) daily protein requirement of ~1.2–1.7 g $\text{kg}^{-1} \text{day}^{-1}$, which is in line with previous recommendations for general athletic populations [31, 32]. Importantly, such nutritional strategies appear to hold functional efficacy: a recent meta-analysis supports a potentiating effect of high-protein feeding upon muscle mass and strength gains afforded by RE programmes [33, 34]. Whilst it is possible to obtain this level and frequency of protein intake through the diet, supplemental protein in various forms including 10 g EAA or 4–5 g leucine [13, 35, 36] are practical methods for ensuring adequate protein ingestion for the concurrent athlete.

CHO Considerations for Both Aerobic and Hypertrophic Conditioning

One of the earliest nutrients to be studied as an ergogenic aid was CHO, the primary performance limiting fuel for endurance-trained individuals exercising at

moderate-high intensities ($\sim 40\text{--}75\%$ $\text{VO}_{2\text{max}}$) lasting 2–3 h [37]. CHO nutritional interventions ultimately aim to manipulate the content of stored CHO in muscle, in the form of glycogen as a metabolic source for glycolysis to drive prolonged muscle contraction. High CHO feeding patterns robustly increase muscle glycogen concentrations [38]. As a result, ‘CHO loading’ ($5\text{--}7\text{ g kg}^{-1}\text{ day}^{-1}$) has been repeatedly shown to enhance performance in endurance exercise lasting $>2\text{ h}$ [39]. Whilst it may be rationalised that CHO loading will maximise the duration/output of individual training sessions, leading to optimal training-induced aerobic adaptation, it is unlikely each (particularly concurrent) training bout will be of a duration (i.e. $>2\text{ h}$) and intensity sufficient to warrant CHO loading. The focus herein will therefore be on how altering dietary CHO and muscle glycogen levels can influence post-exercise adaptation associated with improved aerobic and strength performance following endurance and resistance training, respectively.

Traditionally, endurance training combined with post-exercise CHO ingestion has been promoted for facilitating exercise recovery via expedited glycogen re-synthesis [40] to improve performance during subsequent training bouts. With regard to aerobic conditioning however, high CHO consumption might inhibit several pro-aerobic adaptive signals [41]. Conversely, mounting evidence indicates that limiting muscle CHO availability can enhance endurance training responses. Reflective of higher activation of molecules regulating key aerobic muscle phenotypes, such as mitochondrial biogenesis [42], performing endurance training in the glycogen-depleted state increases oxidative capacity and performance (time to exhaustion) versus a glycogen repleted state [43]. However, despite reporting improved mitochondrial enzyme content/activity and glycogen sparing, subsequent studies have failed to demonstrate performance gains resulting from training under low-glycogen conditions [44–46] or with low-CHO diets [47, 48]. Similarly, despite altering patterns of substrate utilisation, the efficacy of low-CHO, high-fat dieting fails to provide robust improvements in endurance performance or adaptation [reviewed in 37, 49, 50]. Manipulation of fat macronutrient intake is therefore unlikely to yield benefits for concurrent training.

Regardless, an important caveat to integrating glycogen depletion into concurrent training programmes is the impact of low CHO availability upon post-EE protein turnover. Early studies suggested that CHO restriction negatively impacts net protein balance after endurance exercise—primarily via accelerating rates of MPB [51–53]. Future studies directly assessing metabolism using stable isotope tracers extended these findings by highlighting that reduced MPS, as well as the prevailing increases in MPB, contributes to attenuated net protein balance after prolonged exercise when CHO was depleted [54]. Furthermore, although protein and CHO co-ingestion stimulates post-endurance exercise MPS and associated molecular signalling pathways [27], this response is ablated when leucine-enriched feeds are administered in the CHO restricted state [45]. Thus, although EE will increase muscle mass only marginally, if one presumes an appropriate goal for the concurrent athlete is to maximise 24 h net protein accrual to minimise the magnitude of the interference effect, ensuring post-EE CHO re-synthesis can be recommended. For example, in the 0–4 h post-EE period, $\sim 1\text{ g kg body mass}^{-1}$ CHO will suffice, since

glycogen depletion provides a strong drive for its own replenishment. Between 4–24 h post-EE, CHO intake should match the energetic needs of the exercise, at which point total CHO intake is more important than timing or form of CHO [40]. Moderate daily CHO intake ($3\text{--}7\text{ g kg}^{-1}\text{ day}^{-1}$) should suffice for offsetting post-EE decrements in protein balance [55, 56]. This may, however, be at the expense of optimal endurance adaptation.

One further consideration specific to concurrent training is the potential influence of prior exhaustive EE and/or low dietary CHO, and associated muscle glycogen depletion, on the anabolic response to subsequent RE. Reduced glycogen concentrations perturb energy homeostasis to mediate activation of cellular energy-sensing molecules, such as adenosine monophosphate-activated protein kinase (AMPK). Because AMPK is well established to inhibit key regulatory molecules of MPS (e.g. mechanistic target of rapamycin, mTOR) in lower organisms [57, 58], glycogen depletion has been proposed to interfere with the protein synthetic responses to RE [45, 59]. However, this theory has not been substantiated to date. Several studies of pre-RE glycogen depletion or CHO energy restriction consistently highlight that muscle glycogen content has negligible consequences for RE-induced anabolic molecular signalling or MPS [55]. Similarly, supplementing post-RE protein with CHO does not augment the anabolic response to RE [60, 61]. Therefore, despite RE reducing muscle glycogen content by 25–40% [62], EAA present as the principal (and perhaps only) macronutrients required for optimising anabolic responses to exercise [63]. As such, the above observations suggest the high mechanical stimulus associated with RE is sufficient to overcome any putative inhibitory effects of metabolic stress upon muscle growth. Presuming post-RE protein intake and restored energy balance (via standard mixed meal consumption) within ~3 h, manipulation of CHO intake for the RE component of concurrent training is likely to be of low importance. Future research in this area, on the nature of the glycogen-concurrent training relationship will shed important new light on how CHO should be integrated into the diet for facilitating concurrent training.

‘Nutraceuticals’ for Promoting Adaptation to Concurrent Training

The term ‘nutraceuticals’ has recently been proposed to define a nutritional compound that ‘...alone or in tandem with exercise, impacts major physiological end-point(s)’ (e.g. see [64]). In the context of concurrent training, this relates to nutraceuticals that influence metabolism, aerobic capacity and/or muscle size and strength adaptations to EE and RE. However, studies examining specific concurrent training responses to nutraceuticals are even scarcer than for the macronutrients. Knowledge must therefore be sourced again from isolated EE- or RE-induced changes to nutraceutical administration and, additionally, from studies utilising *in vitro* and lower organism (i.e. rodent) models. The following sections will firstly focus on nutraceuticals with potential efficacy for optimising adaptation to both resistance and endurance exercise, under the continued theme of minimising the interference effect. This will be followed by discussion of nutraceuticals with a

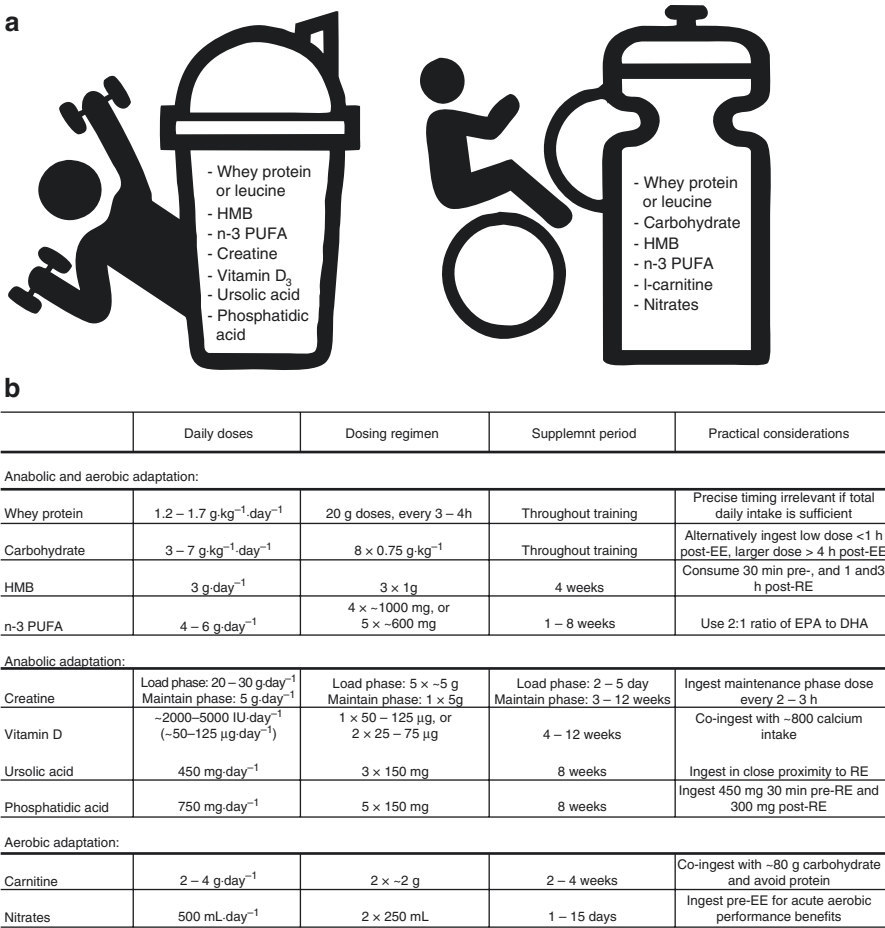


Fig. 16.1 (a) Overview of nutraceuticals that can be considered by concurrent athletes to support resistance (right) and endurance (left) training adaptation. (b) Recommended dosing amounts, protocols and practical considerations for individual nutraceuticals. *HMB* β-hydroxy-β-methylbutyrate, *n-3 PUFA* n-3 polyunsaturated fatty acids, *EPA* eicosapentaenoic acid, *DHA* docosahexaenoic acid, *RE* resistance exercise, *EE* endurance exercise

potential role in augmenting anabolic responses to RE alone, and subsequently aerobic responses to EE (summarised in Fig. 16.1).

Nutraceuticals for Augmenting Anabolic and Aerobic Adaptation to Concurrent Exercise

β-Hydroxy-β-Methylbutyrate (HMB)

The concept that distal metabolites of leucine, the most potent nutrient for stimulating anabolism, was first proposed two decades ago [65]. Leucine can be

metabolised intramuscularly by branched-chain amino acid transaminase to reversibly form α -ketoisocaproate (α -KIC). Downstream metabolism of α -KIC can reach two fates, one of which is to form HMB by α -KIC dioxygenase [66]. Recent work has elucidated the anabolic potential of HMB administered in isolation. In the rested post-absorptive state, 3 g orally ingested HMB stimulated MPS, and inhibited MPB comparably to 3 g leucine [13]. Intriguingly, in contrast to the insulin-mediated inhibition of MPB by leucine, HMB suppressed MPB through insulin independent mechanisms, rendering the attractive possibility that combined leucine-HMB supplementation might maximise fed-state net protein balance by synergistically suppressing MPB.

The acute pro-anabolic feeding effects also appear to potentiate the muscle mass and strength adaptations afforded by RE training. A seminal study [66] reported that 3 g HMB day⁻¹ (vs. 1.5 g HMB day⁻¹ and controls) led to higher gains in muscle strength and lean tissue mass during 3 weeks resistance training. Interestingly, in this study HMB and control groups ingested approximately double the recommended daily intake of protein. This finding is suggestive of additive and/or independent anabolic effects of HMB upon high-protein diets when combined with RE programmes, and is in agreement with recent studies of the acute effects of HMB on rates of MPB [13]. Later works corroborated the efficacy of HMB for facilitating RE-induced muscle strength and mass increases [67, 68]. Because these studies employed untrained [66, 67] and trained [68] volunteers, HMB appears efficacious irrespective of training status. Finally, whilst negligible effects of HMB on responses to resistance training have been reported [69, 70], HMB has been administered in its calcium form (CaHMB). However, since CaHMB has lower bioavailability compared to its free acid form (FA-HMB), CaHMB may result in poorer anabolic potential [71] and FA-HMB may be favoured for sports nutrition.

Importantly, HMB supplementation may also enhance adaptation to EE. For example, 2 weeks of 3 g HMB day⁻¹ in endurance-trained cyclists increased the time to reach VO_{2max} and delayed the onset of blood lactate accumulation, responses indicative of positive aerobic adaptation [72]. Chronic improvements in VO_{2max} [73], aerobic power [74] and physical capacity at the onset of fatigue [75] after 4–5 weeks high-intensity interval training (HIIT) are also potentiated with 3 g HMB day⁻¹ vs. placebo. Additionally, 6 weeks of HMB supplementation attenuated rises in plasma creatine phosphokinase and lactate dehydrogenase after a 20 km time trail run [76], suggesting HMB can attenuate exercise-induced muscle damage (EIMD). Clearly, if such effects persist following RE bouts then HMB might serve as a general nutraceutical for diminishing symptoms of EIMD (e.g. strength decline and sensations of soreness), though this remains to be tested. Whilst the mechanisms of these multiple ergogenic properties are unstudied, HMB is precursor of de novo cholesterol synthesis [65], which is essential for cell membrane maintenance. As such, HMB might preserve membrane integrity during prolonged/muscle damaging exercise.

Collectively, despite a lack of studies directly examining HMB during concurrent training regimes, HMB possesses efficacy for optimising multiple desirable muscular adaptations across exercise modalities, and irrespective of training status. Supplementing 3 g HMB day⁻¹ (ideally in the FA-HMB form) therefore represents

a highly attractive nutraceutical candidate for inclusion into the concurrent athlete's dietary regimen. As a practical consideration, only small amounts of HMB (~5%) are generated from normal leucine metabolism [77], meaning that 60 g leucine would be required to generate the effective HMB dose of 3 g [78]. It is therefore unlikely that HMB is the primary 'active' component of leucine and might act independently and/or synergistically to the anabolic effects of leucine [66]. Thus, HMB supplementation can be considered in addition to standard protein recommendations for the concurrent training un/trained athlete.

n-3 Polyunsaturated Fatty Acids (n-3 PUFAs)

There are three types of n-3 PUFA: (1) alpha-linoleic acid (ALA), (2) eicosapentaenoic acid (EPA) and (3) docosahexaenoic acid (DHA), which can be obtained in the diet with walnuts and oily fish. In untrained individuals, 5 g day⁻¹ fish oil led to continual rises in intramuscular n-3 PUFA concentration throughout the 4 week supplementation period [79]. Parallel increased content of molecular regulators of MPS (e.g. mTOR and focal adhesion kinase) suggests that raised n-3 PUFA positively influences muscle anabolic potential [79]. Indeed, in the fed state (hyperaminoacidaemia-hyperinsulinemia) MPS and associated anabolic signals are potentiated in young healthy individuals following n-3 PUFA (1.86 g day⁻¹ EPA plus 1.5 g day⁻¹ DHA) supplementation for 8 weeks [80]. Individual n-3 PUFAs may also exert distinct anabolic properties. For example, in vitro studies demonstrated upregulation of MPS after EPA treatment, which was absent following DHA [81]; whether this unique efficacy of EPA translates to human is currently unknown.

The efficacy of n-3 PUFA supplementation on the muscular response to RE is less clear. Chronic n-3 PUFA supplementation increases anabolic responses to acute RE [82] and promotes strength gains after 12 weeks resistance training in older individuals [83]. Contrastingly, a single study in younger people found no benefits of 8 weeks fish oil supplementation on the MPS response to an acute bout of RE [84]. However, a lack of pre-supplement measurements renders interpretation of efficacy difficult in this instance. The potential benefits of n-3 PUFA when administered alongside chronic resistance training are also unknown and should be examined in future studies. However, the ergogenic potential of n-3 PUFAs may extend beyond anabolic stimulation. n-3 PUFAs serve as precursors to the prostaglandin family of anti-inflammatory hormones [85], and reduce production of the inflammatory inducer, leukotriene B4 [86]. The role of n-3 PUFAs in aiding post-exercise recovery by attenuating exercise-induced increases in inflammation and perceived muscle soreness has therefore been investigated. Supplementation of 3–6 g day⁻¹ of n-3 PUFA (e.g. 3 g day⁻¹ EPA plus 0.6 g day⁻¹ DHA) for 1–6 weeks prior to an acute bout of strenuous exercise inhibited the appearance of plasma markers of muscle damage, range of motion [87] and muscle soreness at 48 h post-exercise [88, 89]. As such, assuming the normal inflammatory response is not required for appropriate muscle remodelling [90], n-3 PUFA might facilitate training adaptation by expediting the recovery process.

Supplementing n-3 PUFAs can also mediate changes in fuel utilisation during EE. During single bouts of 60–90 min cycling at 60% $\text{VO}_{2\text{max}}$, 3 weeks of prior fish oil intake (6 g day⁻¹) causes higher rates of fat oxidation [91, 92] as a possible compensatory mechanism for lower CHO oxidation. Nonetheless, aerobic adaptive benefits are yet to be determined, and improved fat oxidation rates may have limited efficacy for endurance performance [37]. However, in rodents n-3 PUFA intake upregulates central genes involved in mitochondrial biogenesis, such as peroxisome proliferator-activated receptor- α (PPAR α) [93], and mice fed a low fat, DHA enriched diet displayed higher mitochondrial oxygen consumption vs. placebo—indicative of enhanced mitochondrial function [94]. Overall, therefore, whilst greater clarity of the functional consequences of n-3 PUFAs when combined with RE and/or EE is required, doses of 3–6 g day⁻¹ for periods of 1–6 weeks may contribute to gains in muscle mass and strength after RE, and for augmenting EE-induced aerobic adaptations.

Nutraceuticals for Augmenting Anabolic Adaptation to Resistance Exercise

Creatine

Found almost exclusively in muscle, creatine (Cr) is endogenously synthesised from arginine, glycine and methionine [95]. Creatine can be reversibly phosphorylated by creatine kinase to form phosphocreatine (PCr), whose ‘high-energy’ phosphate can be liberated to rapidly generate ATP from cytosolic ADP. Dietary meat and fish serve as rich sources of exogenous Cr, and are capable of increasing the muscle Cr pool [96]. Alternatively, Cr supplementation of 20–30 g day⁻¹ for ≥ 2 days increases muscle Cr content by >20%, of which ~20–30% is in PCr form [97]. Basal Cr content also determines the efficacy of Cr supplementation, since the greatest gains are observed when Cr levels are low, such as in vegetarians [96].

Whilst Cr supplementation has no negative, nor beneficial effects on EE performance, extensive research has established Cr loading as efficacious for augmenting performance during acute high intensity efforts lasting ~10–30 s [98], and sustaining torque/power output over repeated bouts of sprint exercise [99–101]. These ergogenic benefits are mediated by the Cr-induced rise in muscle PCr, and associated capacity for PCr-based anaerobic ATP re-synthesis both during and after individual sprints, and act predominantly in fatigue-susceptible type II ‘fast-twitch’ muscle fibres [98]. Indeed, studies where muscle Cr levels remained low (~12 mm kg⁻¹ dry weight) found no ergogenic effects of Cr loading on sprint performance [102]. For the concurrent athlete, Cr intake can also influence hypertrophic and functional gains afforded by resistance training. Cr nutrition protocols consisting of a ‘loading phase’ (20–30 g for ~5 days) and subsequent ‘maintenance phase’ (5 g day⁻¹) administered during 12–14 week resistance training programmes leads to greater gains in muscle mass, fibre cross-sectional area and maximal strength capacity vs. placebo [103, 104]. Strikingly, a recent meta-analysis revealed

that Cr was more effective than whey protein when combined with RE training for increasing fat-free mass (1.33 vs. 0.69 kg weight mean difference, respectively) [105]. However, the mechanism(s) by which Cr augments hypertrophic responses to RE remain elusive. Elevated muscle Cr content does not promote MPS either when rested [106] or post-RE [107], but can inhibit resting rates of MPB [108]. Though under debate [109], Cr may act by increasing RE-induced satellite cell activation [110], therein providing the nuclear expansion thought to be required for maintaining DNA content and transcriptional capacity of regenerating/growing muscle fibres. It is also well established that the osmotic shift associated with Cr loading leads to intramuscular water retention and subsequent cell swelling [111]. It is plausible that this cell swelling causes stretch activation of cell membrane-located mediators of muscle anabolism, such as focal adhesion kinase [112]. Finally, enhanced work capacity resulting from Cr loading might increase the stimuli for exercise-induced muscle gene expression changes, thereby augmenting the transcriptional signal for chronic muscle remodelling [112–114]. Regardless of the cellular effectors, supplemental Cr loading protocols would appear to be a stimulus for supporting muscular gains following chronic RE and, although presently unknown, perhaps for maximising muscle hypertrophic gains during concurrent training.

Vitamin D

Vitamin D (VitD) is a steroid hormone obtained primarily through sun exposure, though VitD can be obtained through supplementation or dietary sources (e.g. oily fish, cheese, egg yolks). Following sequential hydroxylation steps at the liver and kidney, circulating VitD is converted to its biologically active form, 25-hydroxyvitamin D, also termed cholecalciferol or VitD₃ [115]. Interest in the effects of VitD on muscle metabolism increased since expression of the VitD receptor (VDR) was observed in skeletal muscle [116], amid conflicting reports of VDR presence in muscle [117, 118] that are likely attributable to discrepant methodologies employed for detecting the receptor [116]. Early in vitro studies provided the pre-clinical efficacy of VitD interventions for increasing myotube size (muscle cells in culture) [119] and increasing the MPS responses to leucine and insulin provision [120]. Although not fully understood, the anabolic action of VitD is thought to occur through two distinct mechanisms. Firstly, elevated intramuscular VitD levels increase VitD binding to the VDR and, consequentially, promote gene transcription via interaction with VitD response elements on the DNA [115, 121]. Secondly, VitD may exert non-genomic effects on secondary messenger signalling pathways. For example, 25-hydroxyvitamin D may bind to the VDR at the cell surface following VDR translocation from the nucleus [122], with subsequent upregulation of Akt, mitogen-activated protein kinases [123] and mTOR complex 1 (mTORC1) [120].

Studies in humans have attempted to forward translate these findings for augmenting muscle mass and function. In doing so it has been shown that VitD supplementation of ~2000–4000 IU day⁻¹ for 4–12 weeks was effective at raising circulating VitD₃ concentrations [124, 125]. The efficacy of such VitD schedules alone on

improving muscle health appears to be confined primarily to elderly individuals [126, 127] where habitual VitD₃ levels are frequently depressed [128]. However, meta-analyses of 30 randomised control trials revealed a small positive impact of VitD supplementation on muscle strength in healthy young adults (in the absence of adjunct exercise), although studies employing all forms of VitD were included in the analyses, without focussing on those that administered VitD₃ alone [128]. It should further be noted that combining VitD₃ with ~800 mg calcium ingestion appears necessary for the strength improving effects of VitD₃ to be observed [124].

A few studies have examined the combined effects of VitD intake and resistance training. A recent systematic review analysed 6 controlled trials that administered VitD (600–5000 IU day⁻¹ for ≥4 weeks) to various athletic populations alongside normal training regimens, which included intermittent team sports (i.e. relevant to concurrent training programmes) [129]. Four studies that employed VitD₃ reported strength improvements that ranged from 1.4 to 18.8%, with 2 studies reporting significant strength improvements ($P < 0.05$). Conversely, 2 studies that supplemented with VitD₂ (a less biologically active form of VitD) had no effect on muscle strength [129]. Furthermore, untrained young men that underwent 4 weeks VitD₃ supplementation (1920 IU day⁻¹) followed by 12 weeks VitD₃ plus resistance training exhibited lower myostatin expression, a negative regulator of muscle mass, and increased the percentage of type IIa muscle fibres [124]. Whilst muscle mass and strength were unaffected by VitD₃ in this study, these morphological changes indicate VitD₃ improves muscle ‘quality’. Supplementing VitD₃ (~2000–5000 IU day⁻¹ for ≥4 weeks) can therefore likely be recommended as part of the concurrent athletes’ nutritional plan.

Phosphatidic Acid

Phosphatidic acid (PA) is a diacyl-glycerophospholipid, which can be sourced in the diet, e.g. from raw cabbage [130]. PA is also present in low amounts endogenously in cell membranes, which can be increased by mitogenic stimulation [131] and RE [132], suggesting a role of PA in muscular responses to growth stimuli. As a lipid second messenger, PA regulates a growing number of cell signalling events [133]. For example, PA treatment of cultured skeletal muscle cells [131] and rodents [134] activates the extracellular signal-related kinase/mTOR anabolic pathway comparably to whey protein. PA interventions also tend to increase rates of MPS, but also blunted the MPS response when combined with whey protein [134]. Administering PA may also benefit net protein balance by inhibiting signals regulating MPB under atrophic conditions [135], although this is far from conclusive.

In humans PA is readily bioavailable, with 1.5 g oral PA sufficient to increase levels of circulating PA (and its extracellular conversion to the predicted bioactive form lysophosphatidic acid) within 30 min and remain elevated for 7 h post-ingestion [136]. When applied to a RE context, 750 mg day⁻¹ PA consumed during 8 weeks supervised resistance training significantly increased lean body mass and quadriceps cross-sectional area vs. placebo, though strength gains were comparable between groups [137]. Similarly, Escalante et al. [138] found augmented muscle

mass and maximal strength gains after 8 weeks RE with daily PA-containing multi-ingredient supplement. However, attributing these positive effects to PA alone is impossible, since the supplement included both HMB and VitD₃. Two further studies employed magnitude-based inference to detect likely to very-likely benefits of PA intake on 8 weeks resistance training-induced lean body mass and strength increases; however, these studies are confounded by a lack of supervised training [139] and lower PA dosages of 250 and 375 mg day⁻¹ [140]. Most recently, Gonzalez et al. [141] employed 750 mg day⁻¹ PA during training protocols consistent with previous studies and found no supplement effect for improving muscle thickness or maximal strength, though body composition was not assessed. Reports on the efficacy of PA for supporting RE adaptation are therefore equivocal and might be explained by methodological discrepancies including: training supervision and design, habitual dietary control or lack thereof, approaches to assessing body composition and performance gains and dosing/timing of PA ingestion [141, 142]. It is also notable that all above studies recruited resistance-trained individuals, thus the effects of PA during the early stages of training where the majority of muscular adaptation occurs [7] are unknown. Nevertheless, the weight of existing evidence supports at least modest gains in muscle size and performance following RE programmes which may facilitate offsetting the interference effect in concurrent programmes. Clearly, further work is needed to define direct ergogenic impacts of RE nutraceuticals in respect to concurrent training.

Nutraceuticals for Augmenting Aerobic Adaptation to Endurance Exercise

Carnitine

Carnitine is endogenously synthesised from AA precursors and exists as D & L isomers; however, only L-carnitine is bioactive [143]. Dietary sources of carnitine include red meat and dairy products. The majority (~95%) of L-carnitine is stored in muscle, the most well established function of which is the translocation of long-chain fatty acids from the cytosol into the mitochondrial matrix for subsequent β -oxidation [144, 145]. This process requires fatty acid conversion to acyl-CoA prior to esterification with carnitine to form acylcarnitine by the mitochondrial membrane enzyme carnitine palmitoyltransferase 1 (CPT1). Acylcarnitine is then transported into the mitochondrial matrix and acts as a substrate for carnitine palmitoyltransferase 1, which converts acylcarnitine to the respective acyl-CoAs for entry into the β -oxidation pathway [146, 147]. As such, without carnitine most dietary lipids cannot be used as energy sources.

L-carnitine has therefore been studied for its ability to enhance fat metabolism and spare glycogen, therein delaying fatigue during prolonged, moderate-high intensity exercise. Acutely elevated muscle carnitine content inhibited CHO oxidation despite high CHO availability, and increased overnight glycogen storage [148]. Similarly, chronic L-carnitine supplementation (1.36–2 g day⁻¹ taken twice daily for 12–24 weeks) reduced glycogen utilisation by ~55% [149] and increased energy expenditure and

CPT1 enzymatic activity [150] during 30 min of low-moderate (50% $\text{VO}_{2\text{max}}$) intensity exercise. Importantly, several studies have demonstrated that L-carnitine loading results in improved endurance performance. Acute intake of 3–15 g L-carnitine ingested 1–2 h pre-exercise improved $\text{VO}_{2\text{max}}$ [151], increased running speed [152] and cycling time to exhaustion [153] in endurance-trained athletes. These findings were accompanied by a stimulation of fat oxidation [153] and decreased lactate appearance [152], suggesting altered fuel metabolism accounted for the performance benefits. Longer-term L-carnitine supplementation in athletes (2–4 g day^{-1} for 2–6 weeks) increased fat oxidation (i.e. lowered respiratory quotient) during prolonged exercise [154], increased $\text{VO}_{2\text{max}}$ by 6% [155] and raised 30 min moderate endurance work output 11% [149]. However, conflicting studies have failed to observe beneficial effects of L-carnitine on fat oxidation [156] and endurance performance [156, 157].

The discrepant findings between studies might be explained by higher exercise intensities employed leading to greater CHO oxidation as a fuel source [149, 158] and failure to include CHO in the L-carnitine dosing regime. The incorporation of CHO into L-carnitine supplementation is pertinent since oral carnitine alone does not increase muscle carnitine stores [159, 160]. Subsequent work discovered that combined carnitine and hyperinsulinaemia increased muscle carnitine accretion ~15% [161]. Practical strategies for overcoming the poor bioavailability of carnitine include co-ingestion with ~80 g CHO [162], owing to the insulinotropic actions of CHO. Conversely, carnitine plus oral protein blunts muscle carnitine uptake [162], though the mechanisms of this phenomenon remain unclear. Overall, 2–4 g day^{-1} of L-carnitine administered acutely or over 2–4 weeks (and ideally in combination with ~80 g CHO) may lead to a shift in metabolism away from CHO and towards fat as a fuel source for low-moderate intensity exercise sessions. The associated improvements in endurance performance capacity may, therefore, maximise the adaptive responses to concurrent training.

Nitrates

Oral ingestion of foods rich in dietary nitrates (NO_3^- , e.g. beetroot and lettuce) results in NO_3^- reduction to nitrite (NO_2^-) by salivary nitrate reductases [163]. The acidic stomach environment subsequently converts NO_2^- to nitric oxide (NO), thus NO_3^- represents an important alternative source of NO to the classical l-arginine-NO synthase pathway [164]. In turn, NO regulates several important functions such as vasodilation, blood flow, glucose/fatty acid oxidation and mitochondrial biogenesis [165]. Supplementing 500 mL day^{-1} of NO_3^- -rich beetroot juice for 6 days [166] or 0.1 mmol day^{-1} of NaNO_3^- for 3 days [167] lowers the O_2 cost of submaximal cycling, indicating improved exercise metabolic efficiency, and increases basal mitochondrial function [168]. Similar reductions in the oxygen cost of exercise have also been reported with longer, 15 days (500 mL day^{-1} beetroot juice) supplementation [169]. Acute consumption of 500 mL NO_3^- between 2.5 and 30 h before exercise also improves 4 and 16 km cycling time trial performance [170] and team sport-specific intermittent performance [171]. However, several authors have failed to observe any ergogenic effect of NO_3^- intake [172–174]. It is likely that the

positive effects of NO_3^- on endurance performance are limited to recreational athletes, since negligible effects of NO_3^- are reported in athletic populations [175, 176]. Higher basal levels of plasma NO_3^- and NO_2^- might partially explain this phenomenon [177]. Supplemental NO_3^- (~500 mL day⁻¹ beetroot juice for 1–15 days) may therefore be recommended to promote endurance performance adaptation in untrained individuals during the initial phases of a concurrent training programme [178], though NO_3^- may become less important once (an as yet unknown) period of muscular adaptation has taken place.

Summary

In the search for the ‘optimal’ exercise nutrition strategy an enormous range of scientific studies has been conducted on the efficacy of a long list of nutritional compounds, often with conflicting or ambiguous results. In the context of minimising the detrimental effects of the interference effect on muscle mass and strength gains, a central component of the concurrent athletes diet should be protein supplementation, ideally administered soon after endurance and resistance exercise bouts, as well as regular high-quality protein-containing meals. At the possible expense of optimal aerobic adaptation, CHO should be ingested post-EE to facilitate glycogen re-synthesis but, importantly, to avoid the catabolic environment associated with low-glycogen levels. Intake of various nutraceuticals may be employed, with precise timings likely being less important than ensuring adequate daily intakes of each respective compound. A practical consideration of note includes ensuring combined carnitine-CHO intake, but avoiding carnitine co-ingestion with protein, to maximise carnitine uptake into muscle. Overall, whilst the efficacy of certain nutraceuticals is contentious, at the very least there is no evidence of nutraceutical supplementation causing detrimental effects to performance, and all nutraceuticals summarised in Fig. 16.1 should be the focus of the concurrent athlete. Finally, future work should aim to provide highly controlled, large cohort and longitudinal studies of the effects specific nutrients have on concurrent training adaptation. Additionally, as concurrent training programmes evolve (e.g. combined HIIT and RE) new nutritional considerations will emerge with a need for new empirical evidence of efficacy.

References

1. Hickson RC. Interference of strength development by simultaneously training for strength and endurance. *Eur J Appl Physiol.* 1980;45:255–63.
2. Coffey VG, Hawley JA. Concurrent exercise training: do opposites distract? *J Physiol.* 2017;595(9):2883–96.
3. Kumar V, Selby A, Rankin D, Patel R, Atherton P, Hildebrandt W, Williams J, Smith K, Seynnes O, Hiscock N, Rennie MJ. Age-related differences in the dose-response of muscle protein synthesis to resistance exercise in young and old men. *J Physiol.* 2009;587:211–7.
4. Biolo G, Maggi SP, Williams BD, Tipton KD, Wolfe RR. Increased rates of muscle protein turnover and amino acid transport after resistance exercise in humans. *Am J Phys.* 1995;268(3 Pt 1):E514–20.

5. Phillips S, Tipton K, Aarsland A, Wolf SE, Wolfe R. Mixed muscle protein synthesis and breakdown after resistance exercise in humans. *Am J Phys.* 1997;273:E99–107.
6. Burd NA, West DWD, Moore DR, Atherton PJ, Staples AW, Prior T, Tang JE, Rennie MJ, Baker SK, Phillips SM. Enhanced amino acid sensitivity of myofibrillar protein synthesis persists for up to 24 h after resistance exercise in young men. *J Nutr.* 2011;141(4):568–73.
7. Brook MS, Wilkinson DJ, Mitchell WK, Lund JN, Szewczyk NJ, Greenhaff PL, Smith K, Atherton PJ. Skeletal muscle hypertrophy adaptations predominate in the early stages of resistance exercise training, matching deuterium oxide-derived measures of muscle protein synthesis and mechanistic target of rapamycin complex 1 signaling. *FASEB J.* 2015;29(11):4485–96.
8. Camera DM, West DW, Phillips SM, Rerich T, Stellingwerff T, Hawley JA, et al. Protein ingestion increases myofibrillar protein synthesis after concurrent exercise. *Med Sci Sports Exerc.* 2015;47(1):82–91.
9. Donges CE, Burd NA, Duffield R, Smith GC, West DW, Short MJ, et al. Concurrent resistance and aerobic exercise stimulates both myofibrillar and mitochondrial protein synthesis in sedentary middle-aged men. *J Appl Physiol.* 2012;112:1992–2001.
10. Smith K, Reynolds N, Downie S, Patel A, Rennie MJ. Effects of flooding amino acids on incorporation of labeled amino acids into human muscle protein. *Am J Phys.* 1998;275:E73–8.
11. Greenhaff PL, Karagounis LG, Peirce N, Simpson EJ, Hazell M, Layfield R, et al. Disassociation between the effects of amino acids and insulin on signaling, ubiquitin ligases, and protein turnover in human muscle. *Am J Physiol Endocrinol Metab.* 2008;295:E595–604.
12. Atherton PJ, Smith K, Etheridge T, Rankin D, Rennie MJ. Distinct anabolic signalling responses to amino acids in C2C12 skeletal muscle cells. *Amino Acids.* 2010a;38:1533–9.
13. Wilkinson DJ, Hossain T, Hill DS, Phillips BE, Crossland H, Williams J, et al. Effects of leucine and its metabolite β -hydroxy- β -methylbutyrate on human skeletal muscle protein metabolism. *J Physiol.* 2013;591:2911–23.
14. Moore DR, Robinson MJ, Fry JL, Tang JE, Glover EI, Wilkinson SB, et al. Ingested protein dose response of muscle and albumin protein synthesis after resistance exercise in young men. *Am J Clin Nutr.* 2009;89:161–8.
15. Witard OC, Jackman SR, Breen L, Smith K, Selby A, Tipton KD. Myofibrillar muscle protein synthesis rates subsequent to a meal in response to increasing doses of whey protein at rest and after resistance exercise. *Am J Clin Nutr.* 2014;99(1):86–95.
16. Tipton KD, Elliott TA, Cree MG, Aarsland AA, Sanford AP, Wolfe RR. Stimulation of net muscle protein synthesis by whey protein ingestion before and after exercise. *Am J Physiol Endocrinol Metab.* 2007;292(1):E71–6.
17. Atherton PJ, Rennie MJ. Protein synthesis a low priority for exercising muscle. *J Physiol.* 2006;573(Pt 2):288–9.
18. Tang JE, Perco JG, Moore DR, Wilkinson SB, Phillips SM. Resistance training alters the response of fed state mixed muscle protein synthesis in young men. *Am J Physiol Regul Integr Comp Physiol.* 2008;294:R172–8.
19. Hoffman JR, Ratamess NA, Tranchina CP, Rashti SL, Kang J, Faigenbaum AD. Effect of protein-supplement timing on strength, power, and body-composition changes in resistance-trained men. *Int J Sport Nutr Exerc Metab.* 2009;19(2):172–85.
20. Schoenfeld BJ, Aragon AA, Krieger JW. The effect of protein timing on muscle strength and hypertrophy: a meta-analysis. *J Int Soc Sports Nutr.* 2013;10(1):53.
21. Atherton PJ, Etheridge T, Watt PW, Wilkinson D, Selby A, Rankin D, et al. Muscle full effect after oral protein: time-dependent concordance and discordance between human muscle protein synthesis and mTORC1 signaling. *Am J Clin Nutr.* 2010b;92:1080–8.
22. Areta JL, Burke LM, Ross ML, Camera DM, West DW, Broad EM, et al. Timing and distribution of protein ingestion during prolonged recovery from resistance exercise alters myofibrillar protein synthesis. *J Physiol.* 2013;591(Pt 9):2319–31.
23. Res PT, Groen B, Pennings B, Beelen M, Wallis GA, Gijsen AP, et al. Protein ingestion before sleep improves postexercise overnight recovery. *Med Sci Sports Exerc.* 2012;44:1560–9.
24. Harber MP, Konopka AR, Jemiolo B, Trappe SW, Trappe TA, Reidy PT. Muscle protein synthesis and gene expression during recovery from aerobic exercise in the fasted and fed states. *Am J Physiol Regul Integr Comp Physiol.* 2010;299(5):R1254–62.

25. Wilkinson SB, Phillips SM, Atherton PJ, Patel R, Yarasheski KE, Tarnopolsky MA, et al. Differential effects of resistance and endurance exercise in the fed state on signalling molecule phosphorylation and protein synthesis in human muscle. *J Physiol.* 2008;586(15):3701–17.
26. Coffey VG, Zhong Z, Shield A, Canny BJ, Chibalin AV, Zierath JR, et al. Early signaling responses to divergent exercise stimuli in skeletal muscle from well-trained humans. *FASEB J.* 2006;20(1):190–2.
27. Breen L, Philp A, Witard OC, Jackman SR, Selby A, Smith K, et al. The influence of carbohydrate-protein co-ingestion following endurance exercise on myofibrillar and mitochondrial protein synthesis. *J Physiol.* 2011;589(Pt 16):4011–25.
28. Moore DR, Camera DM, Areta JL, Hawley JA. Beyond muscle hypertrophy: why dietary protein is important for endurance athletes. *Appl Physiol Nutr Metab.* 2014;7:1–11.
29. Pasiakos SM, McClung HL, McClung JP, Margolis LM, Andersen NE, Cloutier GJ, et al. Leucine-enriched essential amino acid supplementation during moderate steady state exercise enhances postexercise muscle protein synthesis. *Am J Clin Nutr.* 2011;94:809–18.
30. Lunn WR, Pasiakos SM, Colletto MR, Karfonta KE, Carbone JW, Anderson JM, et al. Chocolate milk and endurance exercise recovery: protein balance, glycogen, and performance. *Med Sci Sports Exerc.* 2012;44(4):682–91.
31. Phillips SM. Dietary protein requirements and adaptive advantages in athletes. *Br J Nutr.* 2012;108(Suppl. 2):S158–67.
32. Tarnopolsky M. Protein requirements for endurance athletes. *Nutrition.* 2004;20:662–8.
33. Cermak NM, Res PT, de Groot LC, Saris WH, van Loon LJ. Protein supplementation augments the adaptive response of skeletal muscle to resistance-type exercise training: a meta-analysis. *Am J Clin Nutr.* 2012a;96(6):1454–64.
34. Naclerio F, Larumbe-Zabala E. Effects of whey protein alone or as part of a multi-ingredient formulation on strength, fat-free mass, or lean body mass in resistance-trained individuals: a meta-analysis. *Sports Med.* 2016;46(1):125–37.
35. Atherton PJ, Kumar V, Selby AL, Rankin D, Hildebrandt W, Phillips BE, et al. Enriching a protein drink with leucine augments muscle protein synthesis after resistance exercise in young and older men. *Clin Nutr.* 2017;36(3):888–95.
36. Churchward-Venne TA, Breen L, Di Donato DM, Hector AJ, Mitchell CJ, Moore DR, et al. Leucine supplementation of a low-protein mixed macronutrient beverage enhances myofibrillar protein synthesis in young men: a double-blind, randomized trial. *Am J Clin Nutr.* 2014;99(2):276–86.
37. Hawley JA, Leckey JJ. Carbohydrate dependence during prolonged, intense endurance exercise. *Sports Med.* 2015;45(Suppl 1):S5–12.
38. Hultman E, Bergstrom J. Muscle glycogen synthesis in relation to diet studied in normal subjects. *Acta Med Scand.* 1967;182:109–17.
39. Burke LM, Millet G, Tarnopolsky MA, International Association of Athletics Federations. Nutrition for distance events. *J Sports Sci.* 2007;25(Suppl 1):S29–38.
40. Burke LM, van Loon LJC, Hawley JA. Postexercise muscle glycogen resynthesis in humans. *J Appl Physiol.* 2017;122(5):1055–67.
41. Philp A, Hargreaves M, Baar K. More than a store: regulatory roles for glycogen in skeletal muscle adaptation to exercise. *Am J Physiol Endocrinol Metab.* 2012;302:E1343–51.
42. Philp A, Burke LM, Baar K. Altering endogenous carbohydrate availability to support training adaptations. Nestle Nutr Inst Workshop Ser. 2011;69:19–31.
43. Hansen AK, Fischer CP, Plomgaard P, Andersen JL, Saltin B, Pedersen BK. Skeletal muscle adaptation: training twice every second day vs. training once daily. *J Appl Physiol.* 2005;98:93–9.
44. Hulston CJ, Venables MC, Mann CH, Martin C, Philp A, Baar K, et al. Training with low muscle glycogen enhances fat metabolism in well-trained cyclists. *Med Sci Sports Exerc.* 2010;42:2046–55.
45. Impey SG, Hammond KM, Shepherd SO, Sharples AP, Stewart C, Limb M, et al. Fuel for the work required: a practical approach to amalgamating train-low paradigms for endurance athletes. *Physiol Rep.* 2016;4(10):e12803.
46. Yeo WK, Paton CD, Garnham AP, Burke LM, Carey AL, Hawley JA. Skeletal muscle adaptation and performance responses to once a day versus twice every second day endurance training regimens. *J Appl Physiol.* 2008;105:1462–70.

47. Vogt M, Puntschart A, Howald H, Mueller B, Mannhart C, Gfeller-Tuescher L, et al. Effects of dietary fat on muscle substrates, metabolism, and performance in athletes. *Med Sci Sports Exerc.* 2003;35:952–60.
48. Zajac A, Poprzeczki S, Maszczyk A, Czuba M, Michalczyk M, Zydek G. The effects of a ketogenic diet on exercise metabolism and physical performance in off-road cyclists. *Nutrients.* 2014;6:2493–508.
49. Burke LM. Re-examining high-fat diets for sports performance: did we call the ‘nail in the coffin’ too soon? *Sports Med.* 2015;45(Suppl 1):S33–49.
50. Knuiman P, Hopman MTE, Mensink M. Glycogen availability and skeletal muscle adaptations with endurance and resistance exercise. *Nutr Metab (Lond).* 2015;12:59.
51. Blomstrand E, Saltin B. Effect of muscle glycogen on glucose, lactate and amino acid metabolism during exercise and recovery in human subjects. *J Physiol.* 1999;514(Pt 1):293–302.
52. Lemon PW, Mullin JP. Effect of initial muscle glycogen levels on protein catabolism during exercise. *J Appl Physiol Respir Environ Exerc Physiol.* 1980;48:624–9.
53. Van Hall G, Saltin B, Wagenmakers AJ. Muscle protein degradation and amino acid metabolism during prolonged knee-extensor exercise in humans. *Clin Sci (Lond).* 1999;97:557–67.
54. Howarth KR, Moreau NA, Phillips SM, Gibala MJ. Coingestion of protein with carbohydrate during recovery from endurance exercise stimulates skeletal muscle protein synthesis in humans. *J Appl Physiol.* 2009;106:1394–402.
55. Escobar KA, Van Dusseldorp TA, Kersick CM. Carbohydrate intake and resistance-based exercise: are current recommendations reflective of actual need? *Br J Nutr.* 2017;116:2053–65.
56. Rosset R, Lecoultre V, Egli L, Cros J, Dokumaci AS, Zwygart K, et al. Postexercise repletion of muscle energy stores with fructose or glucose in mixed meals. *Am J Clin Nutr.* 2017;105(3):609–17.
57. Bolster DR, Crozier SJ, Kimball SR, Jefferson LS. AMP-activated protein kinase suppresses protein synthesis in rat skeletal muscle through down-regulated mammalian target of rapamycin (mTOR) signaling. *J Biol Chem.* 2002;277(27):23977–80.
58. Xu J, Ji J, Yan XH. Cross-talk between AMPK and mTOR in regulating energy balance. *Crit Rev Food Sci Nutr.* 2012;52(5):373–81.
59. Pasiakos SM, Vislocky LM, Carbone JW, Altieri N, Konopelski K, Freake HC, et al. Acute energy deprivation affects skeletal muscle protein synthesis and associated intracellular signaling proteins in physically active adults. *J Nutr.* 2010;140:745–51.
60. Hulmi JJ, Laakso M, Mero AA, Häkkinen K, Ahtiainen JP, Peltonen H. The effects of whey protein with or without carbohydrates on resistance training adaptations. *J Int Soc Sports Nutr.* 2015;12:48.
61. Rasmussen BB, Tipton KD, Miller SL, Wolf SE, Wolfe RR. An oral essential amino acid-carbohydrate supplement enhances muscle protein anabolism after resistance exercise. *J Appl Physiol.* 2000;88(2):386–92.
62. Robergs RA, Pearson DR, Costill DL, Fink WJ, Pascoe DD, Benedict MA, et al. Muscle glycogenolysis during differing intensities of weightresistance exercise. *J Appl Physiol.* 1991;70:1700–6.
63. Staples AW, Burd NA, West DW, Currie KD, Atherton PJ, Moore DR, et al. Carbohydrate does not augment exercise-induced protein accretion versus protein alone. *Med Sci Sports Exerc.* 2011;43:1154–61.
64. Deane CS, Wilkinson DJ, Phillips BE, Smith K, Etheridge T, Atherton PJ. “Nutraceuticals” in relation to human skeletal muscle and exercise. *Am J Physiol Endocrinol Metab.* 2017;312:E282–99.
65. Nissen SL, Abumrad NN. Nutritional role of the leucine metabolite β -hydroxy β -methylbutyrate (HMB). *J Nutr Biochem.* 1997;8:300–11.
66. Nissen S, Sharp R, Ray M, Rathmacher JA, Rice D, Fuller JC, et al. Effect of leucine metabolite β -hydroxy- β -methylbutyrate on muscle metabolism during resistance-exercise training. *J Appl Physiol.* 1996;81:2095–104.
67. Panton LB, Rathmacher JA, Baier S, Nissen S. Nutritional supplementation of the leucine metabolite β -hydroxy- β -methylbutyrate (HMB) during resistance training. *Nutrition.* 2000;16:734–9.

68. Wilson JM, Lowery RP, Joy JM, Andersen JC, Wilson SMC, Stout JR, et al. The effects of 12 weeks of beta-hydroxy-beta-methylbutyrate free acid supplementation on muscle mass, strength, and power in resistance-trained individuals: a randomized, double-blind, placebo-controlled study. *Eur J Appl Physiol.* 2014;114:1217–27.
69. Kreider RB, Ferreira M, Wilson M, Almada AL. Effects of calcium β -hydroxy- β methylbutyrate (HMB) supplementation during resistance-training on markers of catabolism, body composition and strength. *Int J Sports Med.* 1999;20:503–9.
70. Slater G, Jenkins D, Logan P, Lee H, Vukovich M, Rathmacher J, et al. Beta hydroxy-beta-methylbutyrate (HMB) supplementation does not affect changes in strength or body composition during resistance training in trained men. *Int J Sport Nutr Exerc Metab.* 2001;11:384–96.
71. Fuller JC, Sharp RL, Angus HF, Khoo PY, Rathmacher JA. Comparison of availability and plasma clearance rates of β -hydroxy- β -methylbutyrate delivery in the free acid and calcium salt forms. *Br J Nutr.* 2015;114:1403–9.
72. Vukovich MD, Dreifort GD. Effect of beta-hydroxy beta-methylbutyrate on the onset of blood lactate accumulation and V(O)₂ peak in endurance-trained cyclists. *J Strength Cond Res.* 2001;15:491–7.
73. Lamboley CRH, Royer D, Dionne JJ. Effects of beta-hydroxy-beta-methylbutyrate on aerobic-performance components and body composition in college students. *Int J Sport Nutr Exerc Metab.* 2007;17:56–69.
74. Robinson EH, Stout JR, Miramonti AA, Fukuda DH, Wang R, Townsend JR, et al. High-intensity interval training and β -hydroxy- β -methylbutyric free acid improves aerobic power and metabolic thresholds. *J Int Soc Sports Nutr.* 2014;11:16.
75. Miramonti AA, Stout JR, Fukuda DH, Robinson EH, Wang R, La Monica MB, et al. The effects of four weeks of high intensity interval training and β -hydroxy- β -methylbutyric free acid supplementation on the onset of neuromuscular fatigue. *J Strength Cond Res.* 2016;30:626–34.
76. Knitter AE, Panton L, Rathmacher JA, Petersen A, Sharp R. Effects of beta-hydroxy-beta-methylbutyrate on muscle damage after a prolonged run. *J Appl Physiol.* 2000;89:1340–4.
77. Van Koeveing M, Nissen S. Oxidation of leucine and α -ketoisocaproate to β -hydroxy- β -methylbutyrate in vivo. *Am J Phys.* 1992;262(1 Pt 1):E27–31.
78. Wilson GJ, Wilson JM, Manninen AH. Effects of beta-hydroxy-beta-methylbutyrate (HMB) on exercise performance and body composition across varying levels of age, sex, and training experience: a review. *Nutr Metab (Lond).* 2008;5(1):17.
79. McGlory C, Galloway SDR, Hamilton DL, McClintock C, Breen L, Dick JR, et al. Temporal changes in human skeletal muscle and blood lipid composition with fish oil supplementation. *Prostaglandins Leukot Essent Fatty Acids.* 2014;90:199–206.
80. Smith GI, Atherton PJ, Reeds DN, Mohammed BS, Rankin D, Rennie MJ, et al. Omega-3 polyunsaturated fatty acids augment the muscle protein anabolic response to hyperaminoacidemia/hyperinsulinemia in healthy young and middle aged men and women. *Clin Sci.* 2011;121:267–78.
81. Kamolrat T, Gray SR. The effect of eicosapentaenoic and docosahexaenoic acid on protein synthesis and breakdown in murine C2C12 myotubes. *Biochem Biophys Res Commun.* 2013;432:593–8.
82. Lalia AZ, Dasari S, Robinson MM, Abid H, Morse DM, Klaus KA, et al. Influence of omega-3 fatty acids on skeletal muscle protein metabolism and mitochondrial bioenergetics in older adults. *Aging.* 2017;9(4):1096–129.
83. Rodacki LF, Pereira G, Naliwaiko K, Coelho I, Pequito D. Fish-oil supplementation enhances the effects of strength training in elderly women. *J Clin Nutr.* 2012;95(2):428–36.
84. McGlory C, Wardle SL, Macnaughton LS, Witard OC, Scott F, Dick J, et al. Fish oil supplementation suppresses resistance exercise and feeding-induced increases in anabolic signaling without affecting myofibrillar protein synthesis in young men. *Physiol Rep.* 2016;4:e12715.
85. Calder PC. n-3 polyunsaturated fatty acids, inflammation, and inflammatory diseases. *Am J Clin Nutr.* 2006;83(6 Suppl):1505S–19S.

86. Weber PC, Fischer D, von Schacky C, Lorenz R, Strasser T. Dietary Omega Polyunsaturated fatty acids and eicosanoid formation in man. In: Health effects of polyunsaturated fatty acids in seafoods. Orlando: Academic Press; 1986. p. 49–60.
87. Corder KE, Newsham KR, McDaniel JL, Ezekiel UR, Weiss EP. Effects of short-term docosahexaenoic acid supplementation on markers of inflammation after eccentric strength exercise in women. *J Sports Sci Med*. 2016;15(1):176–83.
88. Jouris KB, McDaniel JL, Weiss EP. The effect of omega-3 fatty acid supplementation on the inflammatory response to eccentric strength exercise. *J Sport Sci Med*. 2011;10:432–8.
89. Tinsley GM, Gann JJ, Huber SR, Andre TL, La Bounty PM, Bowden RG, et al. Effects of fish oil supplementation on postresistance exercise muscle soreness. *J Diet Suppl*. 2016;21:1–12.
90. Urso ML. Anti-inflammatory interventions and skeletal muscle injury: benefit or detriment? *J Appl Physiol*. 2013;115(6):920–8.
91. Delarue J, Labarthe F, Cohen R. Fish-oil supplementation reduces stimulation of plasma glucose fluxes during exercise in untrained males. *Br J Nutr*. 2003;90:777–86.
92. Huffman DM, Michaelson JL, Thomas TR, Derek M, Huffman JL, Michaelson TRT. Chronic supplementation with fish oil increases fat oxidation during exercise in young men. *J Exerc Physiol*. 2004;7:48–57.
93. Le Guen M, Chaté V, Hiningier-Favier I, Laillet B, Morio B, Pieroni G, et al. A 9-week docosahexaenoic acid-enriched supplementation improves endurance exercise capacity and skeletal muscle mitochondrial function in adult rats. *Am J Physiol Endocrinol Metab*. 2016;310(3):E213–4.
94. Lanza IR, Blachnio-Zabielska A, Johnson ML, Schimke JM, Jakaitis DR, Lebrasseur NK, et al. Influence of fish oil on skeletal muscle mitochondrial energetics and lipid metabolites during high-fat diet. *Am J Physiol Endocrinol Metab*. 2013;304:E1391–403.
95. Bloch K, Schoenheimer R. Biological precursors of creatine. *J Biol Chem*. 1940;138:167–94.
96. Greenhaff PL, Bodin K, Soderlund K, Hultman E. Effect of oral creatine supplementation on skeletal muscle phosphocreatine resynthesis. *Am J Phys*. 1994;266:E725–30.
97. Harris RC, Soderlund K, Hultman E. Elevation of creatine in resting and exercised muscle of normal subjects by creatine supplementation. *Clin Sci (Lond)*. 1992;83:367–74.
98. Casey A, Greenhaff PL. Does dietary creatine supplementation play a role in skeletal muscle metabolism and performance? *Am J Clin Nutr*. 2000;72(2 Suppl):607S–17S.
99. Birch R, Noble D, Greenhaff PL. The influence of dietary creatine supplementation on performance during repeated bouts of maximal isokinetic cycling in man. *Eur J Appl Physiol Occup Physiol*. 1994;69:268–76.
100. Earnest CP, Snell PG, Rodriguez R, Almada AL, Mitchell TL. The effect of creatine monohydrate ingestion on anaerobic power indices, muscular strength and body composition. *Acta Physiol Scand*. 1995;153:207–9.
101. Greenhaff PL, Casey A, Short AH, Harris R, Soderlund K, Hultman E. Influence of oral creatine supplementation of muscle torque during repeated bouts of maximal voluntary exercise in man. *Clin Sci (Lond)*. 1993;84:565–71.
102. Snow RJ, McKenna MJ, Selig SE, Kemp J, Stathis CG, Zhao S. Effect of creatine supplementation on sprint exercise performance and muscle metabolism. *J Appl Physiol*. 1998;84:1667–73.
103. Branch JD. Effect of creatine supplementation on body composition and performance: a meta-analysis. *Int J Sport Nutr Exerc Metab*. 2003;13:198–226.
104. Volek JS, Duncan ND, Mazzetti SA, Staron RS, Putukian M, Gómez AL, et al. Performance and muscle fiber adaptations to creatine supplementation and heavy resistance training. *Med Sci Sports Exerc*. 1999;31:1147–56.
105. Cermak NM, Res PT, de Groot LCPGM, Saris WHM, van Loon LJC. Protein supplementation augments the adaptive response of skeletal muscle to resistance-type exercise training: a meta-analysis. *Am J Clin Nutr*. 2012b;96:1454–64.
106. Louis M, Poortmans JR, Francaux M, Hultman E, Berre J, Boisseau N, et al. Creatine supplementation has no effect on human muscle protein turnover at rest in the postabsorptive or fed states. *Am J Physiol Endocrinol Metab*. 2003a;284:E764–70.

107. Louis M, Poortmans JR, Francaux M, Berré J, Boisseau N, Brassine E, et al. No effect of creatine supplementation on human myofibrillar and sarcoplasmic protein synthesis after resistance exercise. *Am J Physiol Endocrinol Metab*. 2003b;285:E1089–94.
108. Parise G, Mihic S, MacLennan D, Yarasheski KE, Tarnopolsky MA. Effects of acute creatine monohydrate supplementation on leucine kinetics and mixed-muscle protein synthesis. *J Appl Physiol*. 2001;91:1041–7.
109. McCarthy JJ, Esser K. Counterpoint: satellite cell addition is not obligatory for skeletal muscle hypertrophy. *J Appl Physiol*. 2007;103:1100–2.
110. Brook MS, Wilkinson DJ, Phillips BE, Perez-Schindler J, Philp A, Smith K, et al. Skeletal muscle homeostasis and plasticity in youth and ageing: impact of nutrition and exercise. *Acta Physiol*. 2016;216(1):15–41.
111. Ziegenfuss T, Lowery LM, Lemon PW. Acute fluid volume changes in men during three days of creatine supplementation. *J Exerc Physiol*. 1998;1:1–9.
112. Safdar A, Yardley NJ, Snow R, Melov S, Tarnopolsky MA. Global and targeted gene expression and protein content in skeletal muscle of young men following short-term creatine monohydrate supplementation. *Physiol Genomics*. 2008;32:219–28.
113. Buford TW, Kreider RB, Stout JR, Greenwood M, Campbell B, Spano M, et al. International Society of Sports Nutrition position stand: creatine supplementation and exercise. *J Int Soc Sport Nutr*. 2007;4:1–8.
114. Willoughby DS, Rosene J. Effects of oral creatine and resistance training on myosin heavy chain expression. *Med Sci Sports Exerc*. 2001;33:1674–81.
115. Girgis CM, Clifton-Bligh RJ, Hamrick MW, Holick MF, Gunton JE. The roles of vitamin D in skeletal muscle: form, function, and metabolism. *Endocr Rev*. 2013;34:33–83.
116. Girgis CM, Mokbel N, Minn Cha K, Houweling PJ, Abboud M, Fraser DR, et al. The vitamin D receptor (VDR) is expressed in skeletal muscle of male mice and modulates 25-hydroxyvitamin D (25OHD) uptake in myofibers. *Endocrinology*. 2014a;155:3227–37.
117. Pike JW. Expression of the vitamin d receptor in skeletal muscle: are we there yet? *Endocrinology*. 2014;155:3214–8.
118. Wang Y, DeLuca HF. Is the vitamin D receptor found in muscle? *Endocrinology*. 2011;152:354–63.
119. Girgis CM, Clifton-Bligh RJ, Mokbel N, Cheng K, Gunton JE. Vitamin D signaling regulates proliferation, differentiation and myotube size in C2C12 skeletal muscle cells. *Endocrinology*. 2014b;155(2):347–57.
120. Salles J, Chanet A, Giraudet C, Patrac V, Pierre P, Jourdan M, et al. 1,25(OH)₂-vitamin D₃ enhances the stimulating effect of leucine and insulin on protein synthesis rate through Akt/PKB and mTOR mediated pathways in murine C2C12 skeletal myotubes. *Mol Nutr Food Res*. 2013;25:1–10.
121. Ceglia L. Vitamin D and its role in skeletal muscle. *Curr Opin Clin Nutr Metab Care*. 2009;12:628–33.
122. Capiati D, Benassati S, Boland RL. 1,25(OH)₂-vitamin D₃ induces translocation of the vitamin D receptor (VDR) to the plasma membrane in skeletal muscle cells. *J Cell Biochem*. 2002;86:128–35.
123. Buitrago CG, Arango NS, Boland RL. 1 α ,25(OH)₂D₃-dependent modulation of Akt in proliferating and differentiating C2C12 skeletal muscle cells. *J Cell Biochem*. 2012;113:1170–81.
124. Agergaard J, Trøstrup J, Uth J, Iversen JV, Boesen A, Andersen JL, et al. Does vitamin-D intake during resistance training improve the skeletal muscle hypertrophic and strength response in young and elderly men?—a randomized controlled trial. *Nutr Metab (Lond)*. 2015;12:32.
125. Barker T, Schneider ED, Dixon BM, Henriksen VT, Weaver LK. Supplemental vitamin D enhances the recovery in peak isometric force shortly after intense exercise. *Nutr Metab (Lond)*. 2013;10:69.
126. Ceglia L, Niramitmahapanya S, da Silva Morais M, Rivas DA, Harris SS, Bischoff-Ferrari H, et al. A randomized study on the effect of vitamin d₃ supplementation on skeletal muscle morphology and vitamin d receptor concentration in older women. *J Clin Endocrinol Metab*. 2013;98:E1927–35.

127. Ward KA, Das G, Roberts SA, Berry JL, Adams JE, Rawer R, et al. A randomized, controlled trial of vitamin D supplementation upon musculoskeletal health in postmenarchal females. *J Clin Endocrinol Metab.* 2010;95:4643–51.
128. Beaudart C, Buckinx F, Rabenda V, Gillain S, Cavalier E, Slomian J, et al. The effects of vitamin D on skeletal muscle strength, muscle mass and muscle power: a systematic review and meta-analysis of randomized controlled trials. *J Clin Endocrinol Metab.* 2014;99:4336–45.
129. Chiang CM, Ismaeel A, Griffis RB, Weems S. Effects of vitamin D supplementation on muscle strength in athletes: a systematic review. *J Strength Cond Res.* 2017;31(2):566–74.
130. Tanaka T, Kassai A, Ohmoto M, Morito K, Kashiwada Y, Takaishi Y, et al. Quantification of phosphatidic acid in foodstuffs using a thin-layer-chromatography-imaging technique. *J Agric Food Chem.* 2012;60:4156–61.
131. Fang Y, Vilella-Bach M, Bachmann R, Flanigan A, Chen J. Phosphatidic acid-mediated mitogenic activation of mTOR signaling. *Science.* 2001;294:1942–5.
132. Rasmussen BB. Phosphatidic acid: a novel mechanical mechanism for how resistance exercise activates mTORC1 signalling. *J Physiol.* 2009;587(Pt 14):3415–6.
133. English D, Cui Y, Siddiqui RA. Messenger functions of phosphatidic acid. *Chem Phys Lipids.* 1996;80:117–32.
134. Mobley CB, Hornberger TA, Fox CD, Healy JC, Ferguson BS, Lowery RP, et al. Effects of oral phosphatidic acid feeding with or without whey protein on muscle protein synthesis and anabolic signaling in rodent skeletal muscle. *J Int Soc Sports Nutr.* 2015;12:32.
135. Jaafar R, De Larichaudy J, Chanon S, Euthine V, Durand C, Naro F, et al. Phospholipase D regulates the size of skeletal muscle cells through the activation of mTOR signaling. *Cell Commun Signal.* 2013;11:55.
136. Purpura M, Jäger R, Joy JM, Lowery RP, Moore JD, Wilson JM. Effect of oral administration of soy-derived phosphatidic acid on concentrations of phosphatidic acid and lyso-phosphatidic acid molecular species in human plasma. *J Int Soc Sports Nutr.* 2013;10:P22.
137. Joy JM, Gundermann DM, Lowery RP, Jäger R, McCleary SA, Purpura M, et al. Phosphatidic acid enhances mtor signaling and resistance exercise induced hypertrophy. *Nutr Metab.* 2014;11:1.
138. Escalante G, Alencar M, Haddock B, Harvey P. The effects of phosphatidic acid supplementation on strength, body composition, muscular endurance, power, agility, and vertical jump in resistance trained men. *J Int Soc Sports Nutr.* 2016;13(1):1.
139. Hoffman JR, Stout JR, Williams DR, Wells AJ, Fragala MS, Mangine GT, et al. Efficacy of phosphatidic acid ingestion on lean body mass, muscle thickness and strength gains in resistance-trained men. *J Int Soc Sports Nutr.* 2012;9(1):1.
140. Andre TL, Gann JJ, McKinley-Barnard SK, Song JJ, Willoughby DS. Eight weeks of phosphatidic acid supplementation in conjunction with resistance training does not differentially affect body composition and muscle strength in resistance-trained men. *J Sports Sci Med.* 2016;15:532–9.
141. Gonzalez AM, Sell KM, Ghigiarelli JJ, Kelly CF, Shone EW, Accetta MR, et al. Effects of phosphatidic acid supplementation on muscle thickness and strength in resistance-trained men. *Appl Physiol Nutr Metab.* 2017;42(4):443–8.
142. Bond P. Phosphatidic acid: biosynthesis, pharmacokinetics, mechanisms of action and effect on strength and body composition in resistance-trained individuals. *Nutr Metab (Lond).* 2017;14:12.
143. Bremer J. Carnitine—metabolism and functions. *Physiol Rev.* 1983;63:1420–80.
144. Fritz IB, Marquis NR. The role of acylcarnitine esters and carnitine palmityltransferase in the transport of fatty acyl groups across mitochondrial membranes. *Proc Natl Acad Sci U S A.* 1965;54:1226–33.
145. Kerner J, Hoppel C. Fatty acid transport into mitochondria. *Biochem Biophys Acta.* 2000;1486:1–17.
146. Hoppel C. The role of carnitine in normal and altered fatty acid metabolism. *Am J Kidney Dis.* 2003;41:4–12.

147. Parvin R, Pande SV. Enhancement of mitochondrial carnitine and carnitine acylcarnitine translocase-mediated transport of fatty acids into liver mitochondria under ketogenic conditions. *J Biol Chem.* 1979;245:5423–9.
148. Stephens FB, Constantin-Teodosiu D, Laithwaite D, Simpson EJ, Greenhaff PL. Insulin stimulates L-carnitine accumulation in human skeletal muscle. *FASEB J.* 2006a;20:377–9.
149. Wall BT, Stephens FB, Constantin-Teodosiu D, Marimuthu K, Macdonald IA, Greenhaff PL. Chronic oral ingestion of L-carnitine and carbohydrate increases muscle carnitine content and alters muscle fuel metabolism during exercise in humans. *J Physiol.* 2011;589:963–73.
150. Stephens FB, Wall BT, Marimuthu K, Shannon CE, Constantin-Teodosiu D, Macdonald IA, et al. Skeletal muscle carnitine loading increases energy expenditure, modulates fuel metabolism gene networks and prevents body fat accumulation in humans. *J Physiol.* 2013;591(18):4655–66.
151. Vecchiet L, Di Lisa F, Pieralisi G, Ripari P, Menabò R, Giamberardino MA, et al. Influence of L-carnitine administration on maximal physical exercise. *Eur J Appl Physiol Occup Physiol.* 1990;61(5-6):486–90.
152. Ozer GE, Guzel NA. The effects of acute L-carnitine supplementation on endurance performance of athletes. *J Strength Cond Res.* 2014;28(2):514–9.
153. Cha YS, Choi SK, Suh H, Lee SN, Cho D, Li K. Effects of carnitine coingested caffeine on carnitine metabolism and endurance capacity in athletes. *J Nutr Sci Vitaminol (Tokyo).* 2001;47(6):378–84.
154. Gorostiaga EM, Maurer CA, Eclache JP. Decrease in respiratory quotient during exercise following L-carnitine supplementation. *Int J Sports Med.* 1989;10(3):169–74.
155. Marconi C, Sassi G, Carpinelli A, Cerretelli P. Effects of L-carnitine loading on the aerobic and anaerobic performance of endurance athletes. *Eur J Appl Physiol Occup Physiol.* 1985;54(2):131–5.
156. Broad EM, Maughan RJ, Galloway SD. Effects of four weeks L-carnitine L-tartrate ingestion on substrate utilization during prolonged exercise. *Int J Sport Nutr Exerc Metab.* 2005;15(6):665–79.
157. Lee JK, Lee JS, Park H, Cha YS, Yoon CS, Kim CK. Effect of L-carnitine supplementation and aerobic training on FABPc content and beta-HAD activity in human skeletal muscle. *Eur J Appl Physiol.* 2007;99(2):193–9.
158. Gonzalez JT, Stevenson EJ. New perspectives on nutritional interventions to augment lipid utilisation during exercise. *Br J Nutr.* 2012;107:339–49.
159. Soop M, Björkman O, Cederblad G, Hagenfeldt L, Wahren J. Influence of carnitine supplementation on muscle substrate and carnitine metabolism during exercise. *J Appl Physiol.* 1988;64:2394–9.
160. Wächter S, Vogt M, Kreis R, Boesch C, Bigler P, Hoppeler H, et al. Long-term administration of L-carnitine to humans: effect on skeletal muscle carnitine content and physical performance. *Clin Chim Acta.* 2002;318:51–61.
161. Stephens FB, Constantin-Teodosiu D, Laithwaite D, Simpson EJ, Greenhaff PL. An acute increase in skeletal muscle carnitine content alters fuel metabolism in resting human skeletal muscle. *J Clin Endocrinol Metab.* 2006b;91:5013–8.
162. Shannon CE, Nixon AV, Greenhaff PL, Stephens FB. Protein ingestion acutely inhibits insulin-stimulated muscle carnitine uptake in healthy young men. *Am J Clin Nutr.* 2016;103:276–82.
163. Duncan C, Dougall H, Johnston P, Green S, Brogan R, Leifert C, et al. Chemical generation of nitric oxide in the mouth from the enterosalivary circulation of dietary nitrate. *Nat Med.* 1995;1:546–51.
164. Lundberg JO, Weitzberg E, Gladwin MT. The nitrate-nitrite-nitric oxide pathway in physiology and therapeutics. *Nat Rev Drug Discov.* 2008;7(2):156–67.
165. Stamler JS, Meissner G. Physiology of nitric oxide in skeletal muscle. *Physiol Rev.* 2001;81(1):209–37.
166. Bailey SJ, Winyard P, Vanhatalo A, Blackwell JR, Dimenna FJ, Wilkerson DP, et al. Dietary nitrate supplementation reduces the O₂ cost of low-intensity exercise and enhances tolerance to high-intensity exercise in humans. *J Appl Physiol.* 2009;107:1144–55.

167. Larsen FJ, Weitzberg E, Lundberg JO, Ekblom B. Effects of dietary nitrate on oxygen cost during exercise. *Acta Physiol.* 2007;191:59–66.
168. Larsen FJ, Schiffer TA, Borniquel S, Sahlin K, Ekblom B, Lundberg JO, et al. Dietary inorganic nitrate improves mitochondrial efficiency in humans. *Cell Metab.* 2011;13:149–59.
169. Vanhatalo A, Bailey SJ, Blackwell JR, DiMenna FJ, Pavey TG, Wilkerson DP, et al. Acute and chronic effects of dietary nitrate supplementation on blood pressure and the physiological responses to moderate-intensity and incremental exercise. *Am J Physiol Regul Integr Comp Physiol.* 2010;299(4):R1121–31.
170. Lansley KE, Winyard PG, Bailey SJ, Vanhatalo A, Wilkerson DP, Blackwell JR, et al. Acute dietary nitrate supplementation improves cycling time trial performance. *Med Sci Sports Exerc.* 2011;43:1125–31.
171. Wylie LJ, Mohr M, Krstrup P, Jackman SR, Ermidis G, Kelly J, et al. Dietary nitrate supplementation improves team sport-specific intense intermittent exercise performance. *Eur J Appl Physiol.* 2013;113:1673–84.
172. Arnold JT, Oliver SJ, Lewis-Jones TM, Wylie LJ, Macdonald JH. Beetroot juice does not enhance altitude running performance in well-trained athletes. *Appl Physiol Nutr Metab.* 2015;40:590–5.
173. Cermak NM, Res P, Stinkens R, Lundberg JO, Gibala MJ, van Loon LJ. No improvement in endurance performance after a single dose of beetroot juice. *Int J Sport Nutr Exerc Metab.* 2012c;22:470–8.
174. Wilkerson DP, Hayward GM, Bailey SJ, Vanhatalo A, Blackwell JR, Jones AM. Influence of acute dietary nitrate supplementation on 50 mile time trial performance in well-trained cyclists. *Eur J Appl Physiol.* 2012;112:4127–34.
175. Jonvik KL, Nyakayiru J, van Loon LJ, Verdijk LB. Can elite athletes benefit from dietary nitrate supplementation? *J Appl Physiol.* 2015;119(6):759–61.
176. Poortmans JR, Gualano B, Carpentier A. Nitrate supplementation and human exercise performance: too much of a good thing? *Curr Opin Clin Nutr Metab Care.* 2015;18(6):599–604.
177. Poveda JJ, Riestra A, Salas E, Cagigas ML, López-Somoza C, Amado JA, et al. Contribution of nitric oxide to exercise-induced changes in healthy volunteers: effects of acute exercise and long-term physical training. *Eur J Clin Invest.* 1997;27:967–71.
178. Perez-Schindler J, Hamilton LD, Moore DR, Baar K, Philp A. Nutritional strategies to support concurrent training. *Eur J Sport Sci.* 2015;15(1):41–52.